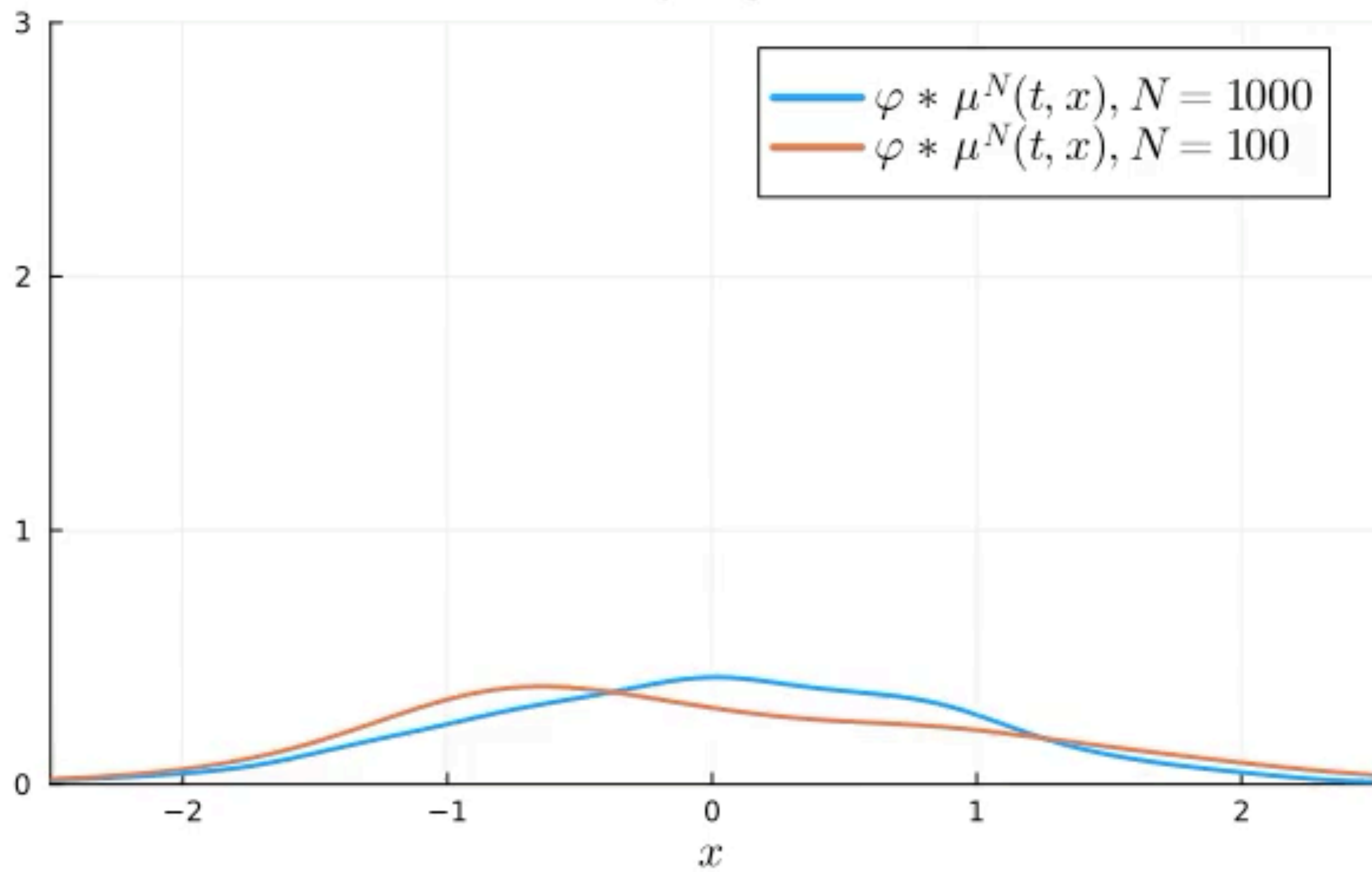


AMKean-Vasovmodel

- $$dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)dt + \sigma dW_t^i$$

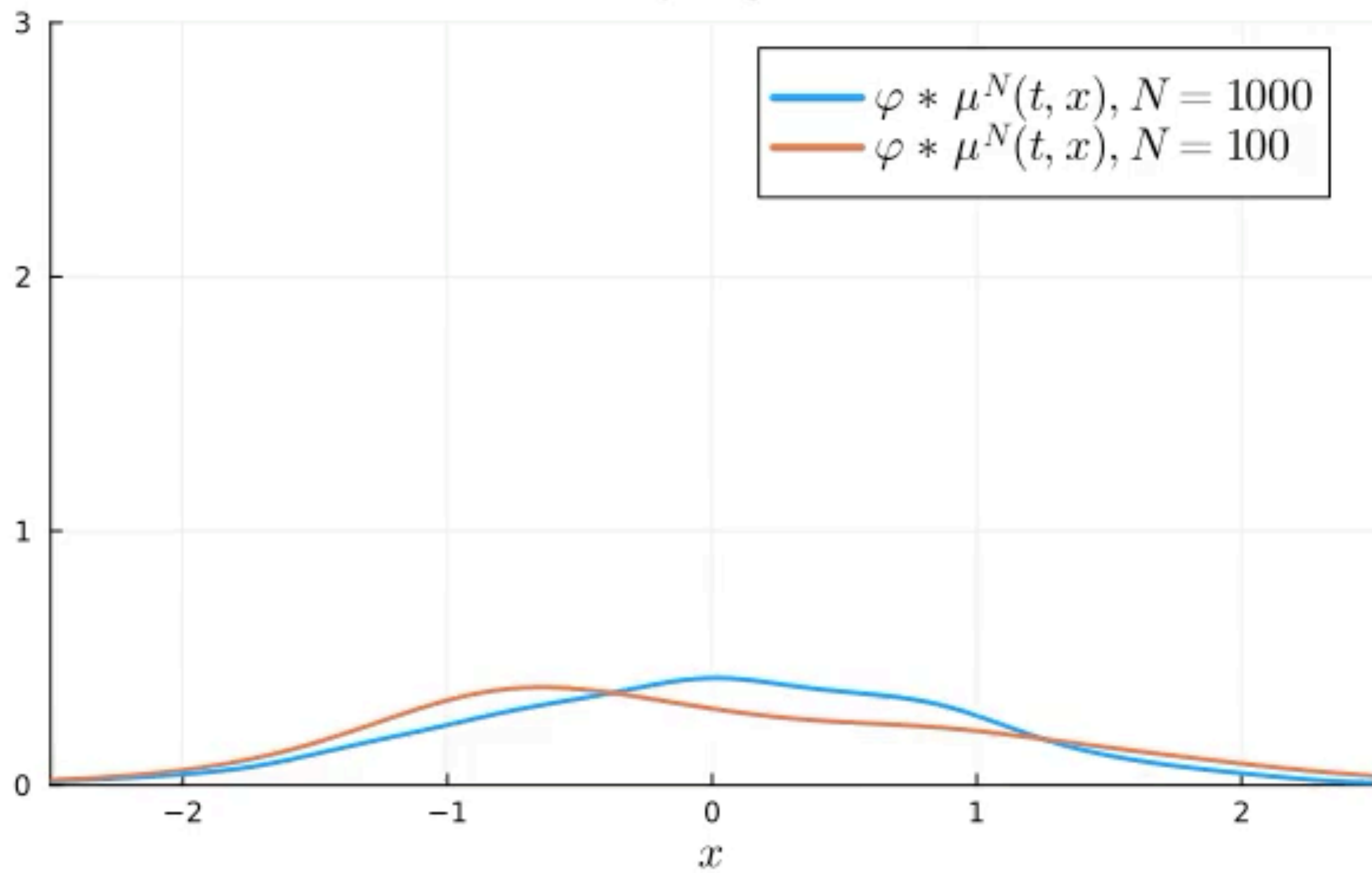
- Empirical measure
$$\mu_t^N = \frac{1}{N} \sum_i \delta_{X_t^i}$$

$t = 0$



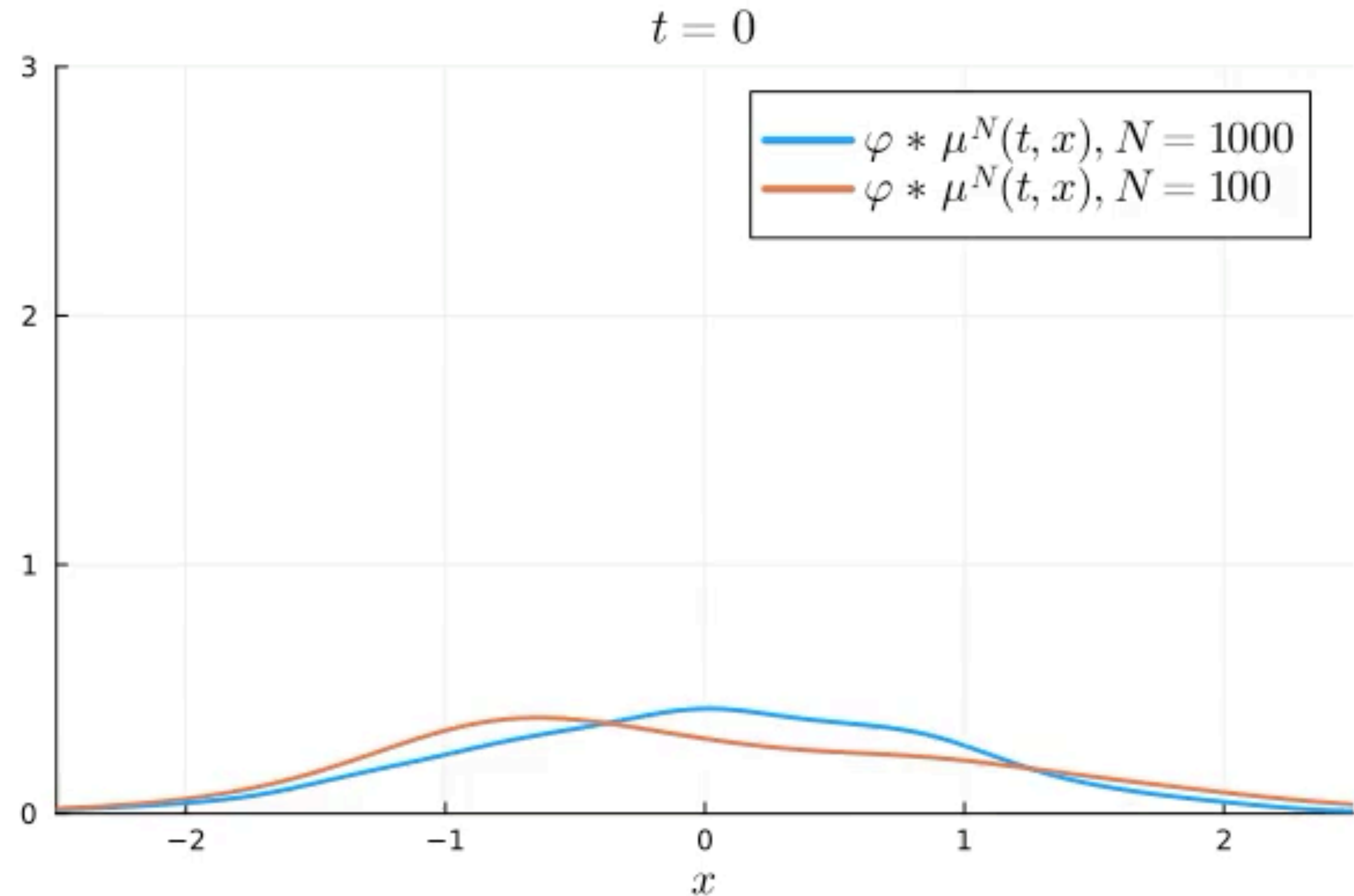
$$a \equiv -1/N, \sigma \equiv 0.1$$

$t = 0$



A McKean-Vlasov model

- $dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)dt + \sigma dW_t^i$
- Empirical measure $\mu_t^N = \frac{1}{N} \sum_i \delta_{X_t^i}$



$$a = -1/N, \sigma = 0.1$$

McKean-Vlasov models

Theory (1960-1990: McKean, Hitsuda, Méléard, ...)

$$\left\{ \mathrm{d}X_t^i = b(X_t^i, \mu_t^N) \mathrm{d}t + \sqrt{2\sigma} \mathrm{d}W_t^i \right.$$

- Mean-field scaling: $a = O(1/N)$
- $\mu_t^N \rightarrow \mathcal{L}(X_t) \hat{=} \mu_t$ in law as $N \rightarrow \infty$
- μ_t solves McKean-Vlasov PDE $\partial_t \mu_t = \nabla_x \cdot (\sigma \nabla_x \mu_t - b(x, \mu_t) \mu_t)$
- $\sqrt{N}(\mu_t^N - \mu_t)$ converges to a Gaussian measure-valued process